# Laboratory 4

# Variant #2

Regression and Classification

Diabetes Disease Progression Prediction

Deniz Yilmaz

Introduction to Artificial Intelligence

Warsaw University of Technology, Summer 2026

## 1.0 Introduction

The goal of this laboratory task was to solve a regression problem using the diabetes dataset. The task is to predict disease progression based on ten numerical medical measurements. Since the target value is continuous, the problem is regression rather than classification.-

The dataset contains 442 samples. The input columns are age, sex, bmi, bp, s1, s2, s3, s4, s5, and s6. The target column stores the disease progression value that the models try to predict. The main evaluation metric used in this report is Root Mean Squared Error (RMSE), where a lower value means better predictions.

----------------------------------------------------------------------------------------------------

## 2.0 Data Preparation and Implementation

The dataset was loaded in Python and prepared before training. The column names were printed and checked. Rows with missing values were removed using dropna(), and duplicate rows were removed using drop_duplicates(). No feature column was removed because the dataset does not contain an obvious directly dependent feature, such as a column that is only the sum of other columns.

The target column was separated from the input features. The input features were split into training and testing sets using an 80/20 split with random_state = 42. Feature scaling was applied using StandardScaler because KNN is distance-based. Without scaling, features with larger numerical ranges could have too much influence on the distance calculation.

### 2.1 Model Choice and Hyperparameters

Two regression models were used. The first model was K-Nearest Neighbors Regressor. KNN is a lazy learning method, meaning that training mainly stores the training samples. During prediction, the model finds the closest training samples and predicts from their target values.

The second model was Linear Regression. It was used as a simple baseline model because it learns a linear relationship between the input features and the target value.

## 3.0 Experiments

The KNN model was tested using different odd k values from 3 to 41. For each value of k, the model was trained on the training set and evaluated on the test set using RMSE. The distance metric was selected using GridSearchCV with 4-fold cross-validation. The final recorded results are shown in Table 1.

Table 1: KNN test RMSE for different k values

| k value | RMSE |
| --- | --- |
| 3 | 57.2816 |
| 5 | 54.0312 |
| 7 | 54.4933 |
| 9 | 55.4163 |
| 11 | 55.6659 |
| 13 | 55.7337 |
| 15 | 55.2610 |
| 17 | 54.5478 |
| 19 | 54.8667 |
| 21 | 55.3435 |
| 23 | 55.1601 |
| 25 | 54.6948 |
| 27 | 54.7249 |
| 29 | 55.4784 |
| 31 | 55.3982 |
| 33 | 55.6453 |
| 35 | 55.4330 |
| 37 | 55.4712 |

**Best KNN result:** k = 5, RMSE = 54.0312.

**Linear Regression result:** RMSE = 53.8534.

**Best overall result:** Linear Regression gave the lowest RMSE, with a difference of 0.1777 compared with the best KNN result.

## 3.0 Conclusion

In conclusion, both KNN Regression and Linear Regression were able to predict diabetes disease progression with similar performance. The best KNN model used k = 5 and achieved an RMSE of 54.0312. Linear Regression achieved a slightly better RMSE of 53.8534.

The results suggest that a simple linear model is slightly more effective for this dataset than the tested KNN configurations. KNN still performed reasonably well, but it was sensitive to the selected value of k. The experiment also shows why normalization is important for KNN, since its predictions are based on distances between samples. Overall, Linear Regression was the best model in these experiments, while KNN provided a useful comparison as a lazy distance-based regression method.

## 4.0 Potential Improvements

Different seeds can be used to extend number of experiements.

Other models can be implemented to compare with.

A lazy comparison like knn might have not been the best option.